

5. (a) Given that

$$y = \arcsin x \quad -1 \leq x \leq 1$$

show that

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}} \quad (3)$$

(b) $f(x) = \arcsin(e^x) \quad x \leq 0$

Prove that $f(x)$ has no stationary points.

(3)

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5. The curve C has equation

$$y = \arccos\left(\frac{1}{2}x\right) \quad -2 \leq x \leq 2$$

(a) Show that C has no stationary points.

(3)

The normal to C , at the point where $x = 1$, crosses the x -axis at the point A and crosses the y -axis at the point B .

Given that O is the origin,

(b) show that the area of the triangle OAB is $\frac{1}{54}(p\sqrt{3} + q\pi + r\sqrt{3}\pi^2)$ where p , q and r are integers to be determined.

(5)

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5. (a)

$$y = \tan^{-1} x$$

Assuming the derivative of $\tan x$, prove that

$$\frac{dy}{dx} = \frac{1}{1+x^2}$$

(3)

$$f(x) = x \tan^{-1} 4x$$

(b) Show that

$$\int f(x) dx = Ax^2 \tan^{-1} 4x + Bx + C \tan^{-1} 4x + k$$

where k is an arbitrary constant and A , B and C are constants to be determined.

(5)

(c) Hence find, in exact form, the mean value of $f(x)$ over the interval $\left[0, \frac{\sqrt{3}}{4}\right]$

(2)

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2. (a) Write $x^2 + 4x - 5$ in the form $(x + p)^2 + q$ where p and q are integers.

(1)

(b) Hence use a standard integral from the formula book to find

$$\int \frac{1}{\sqrt{x^2 + 4x - 5}} dx$$

(2)

(c) Determine the mean value of the function

$$f(x) = \frac{1}{\sqrt{x^2 + 4x - 5}} \quad 3 \leq x \leq 13$$

giving your answer in the form $A \ln B$ where A and B are constants in simplest form.

(3)

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3.

$$f(x) = \frac{1}{\sqrt{4x^2 + 9}}$$

- (a) Using a substitution, that should be stated clearly, show that

$$\int f(x)dx = A \sinh^{-1}(Bx) + c$$

where c is an arbitrary constant and A and B are constants to be found.

(4)

- (b) Hence find, in exact form in terms of natural logarithms, the mean value of $f(x)$ over the interval $[0, 3]$.

(2)

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6. (a) Express as partial fractions

$$\frac{2x^2 + 3x + 6}{(x+1)(x^2 + 4)}$$

(3)

(b) Hence, show that

$$\int_0^2 \frac{2x^2 + 3x + 6}{(x+1)(x^2 + 4)} dx = \ln(a\sqrt{2}) + b\pi$$

where a and b are constants to be determined.

(4)

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3. (a) Explain why

$$\int_{\frac{4}{3}}^{\infty} \frac{1}{9x^2 + 16} dx$$

is an improper integral.

(1)

(b) Show that

$$\int_{\frac{4}{3}}^{\infty} \frac{1}{9x^2 + 16} dx = k\pi$$

where k is a constant to be determined.

(4)



2. (a) Explain why $\int_1^{\infty} \frac{1}{x(2x+5)} dx$ is an improper integral. (1)

(b) Prove that

$$\int_1^{\infty} \frac{1}{x(2x+5)} dx = a \ln b$$

where a and b are rational numbers to be determined.

(6)

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5. (i) Evaluate the improper integral

$$\int_1^{\infty} 2e^{-\frac{1}{2}x} dx \quad (3)$$

- (ii) The air temperature, $\theta^\circ\text{C}$, on a particular day in London is modelled by the equation

$$\theta = 8 - 5 \sin\left(\frac{\pi}{12}t\right) - \cos\left(\frac{\pi}{6}t\right) \quad 0 \leq t \leq 24$$

where t is the number of hours after midnight.

- (a) Use calculus to show that the mean air temperature on this day is 8°C , according to the model. (3)

Given that the actual mean air temperature recorded on this day was higher than 8°C ,

- (b) explain how the model could be refined. (1)

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2. Show that

$$\int_0^{\infty} \frac{8x - 12}{(2x^2 + 3)(x + 1)} dx = \ln k$$

where k is a rational number to be found.

(7)

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9. (i) (a) Explain why $\int_0^{\infty} \cosh x \, dx$ is an improper integral.

(1)

(b) Show that $\int_0^{\infty} \cosh x \, dx$ is divergent.

(3)

(ii) $4 \sinh x = p \cosh x$ where p is a real constant

Given that this equation has real solutions, determine the range of possible values for p

(2)

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9. (a) Use a hyperbolic substitution and calculus to show that

$$\int \frac{x^2}{\sqrt{x^2 - 1}} dx = \frac{1}{2} \left[x\sqrt{x^2 - 1} + \operatorname{arcosh} x \right] + k$$

where k is an arbitrary constant.

(6)

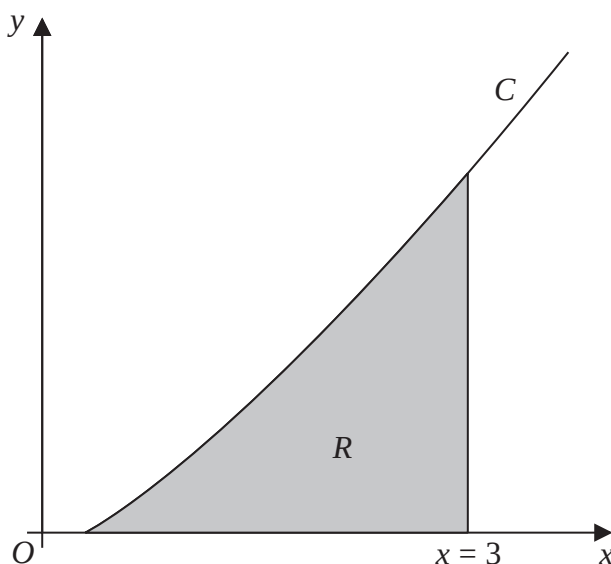


Figure 1

Figure 1 shows a sketch of part of the curve C with equation

$$y = \frac{4}{15} x \operatorname{arcosh} x \quad x \geq 1$$

The finite region R , shown shaded in Figure 1, is bounded by the curve C , the x -axis and the line with equation $x = 3$

- (b) Using algebraic integration and the result from part (a), show that the area of R is given by

$$\frac{1}{15} \left[17 \ln(3 + 2\sqrt{2}) - 6\sqrt{2} \right]$$

(5)



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9.

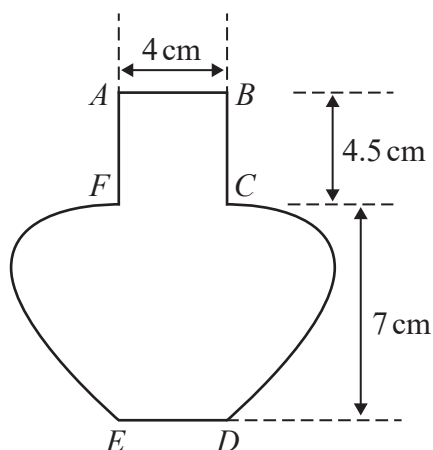


Figure 1

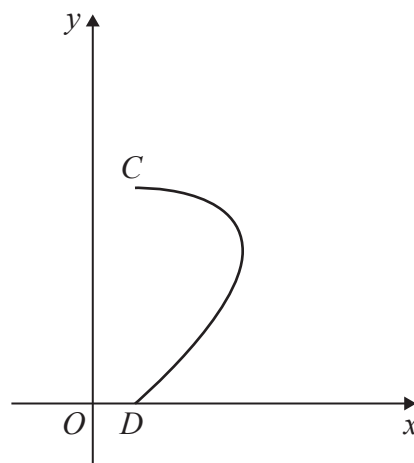


Figure 2

Figure 1 shows the central vertical cross-section $ABCDEFA$ of a vase together with measurements that have been taken from the vase.

The horizontal cross-section between AB and FC is a circle with diameter 4 cm.

The base of the vase ED is horizontal and the point E is vertically below F and the point D is vertically below C .

Using these measurements, the curve CD is modelled by the parametric equations

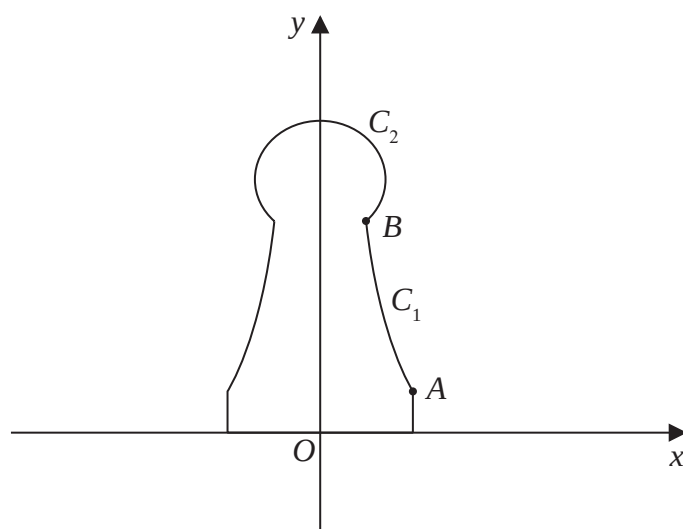
$$x = a + 3 \sin 2t \quad y = b \cos t \quad 0 \leq t \leq \frac{\pi}{2}$$

where a and b are constants and O is the fixed origin, as shown in Figure 2.

- (a) Determine the value of a and the value of b according to the model. (2)
- (b) Using algebraic integration and showing all your working, determine, according to the model, the volume of the vase, giving your answer to the nearest cm^3 (7)
- (c) State a limitation of the model. (1)



7.

**Figure 1**

A student wants to make plastic chess pieces using a 3D printer. Figure 1 shows the central vertical cross-section of the student's design for one chess piece. The plastic chess piece is formed by rotating the region bounded by the y -axis, the x -axis, the line with equation $x = 1$, the curve C_1 and the curve C_2 through 360° about the y -axis.

The point A has coordinates $(1, 0.5)$ and the point B has coordinates $(0.5, 2.5)$ where the units are centimetres.

The curve C_1 is modelled by the equation

$$x = \frac{a}{y + b} \quad 0.5 \leq y \leq 2.5$$

(a) Determine the value of a and the value of b according to the model.

(2)

The curve C_2 is modelled to be an arc of the circle with centre $(0, 3)$.

(b) Use calculus to determine the volume of plastic required to make the chess piece according to the model.

(9)

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7.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

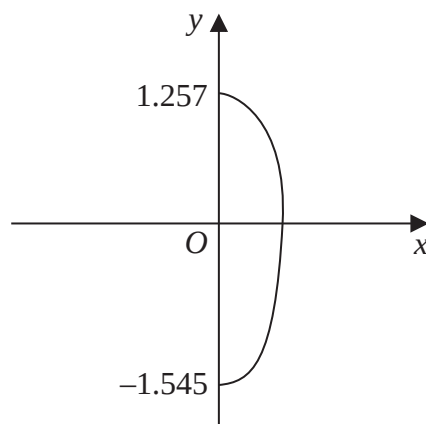


Figure 2

John picked 100 berries from a plant.

The largest berry picked was approximately 2.8 cm long.

The shape of this berry is modelled by rotating the curve with equation

$$16x^2 + 3y^2 - y \cos\left(\frac{5}{2}y\right) = 6 \quad x \geq 0$$

shown in Figure 2, about the y-axis through 2π radians, where the units are cm.

Given that the y intercepts of the curve are -1.545 and 1.257 to four significant figures,

- (a) use algebraic integration to determine, according to the model, the volume of this berry.

(6)

Given that the 100 berries John picked were then squeezed for juice,

- (b) use your answer to part (a) to decide whether, in reality, there is likely to be enough juice to fill a 200 cm^3 cup, giving a reason for your answer.

(2)



8. (a) Given

$$z^n + \frac{1}{z^n} = 2 \cos n\theta \quad n \in \mathbb{N}$$

show that

$$32 \cos^6 \theta \equiv \cos 6\theta + 6 \cos 4\theta + 15 \cos 2\theta + 10 \quad (5)$$

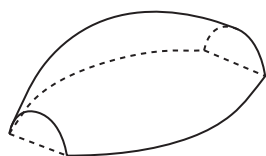


Figure 1

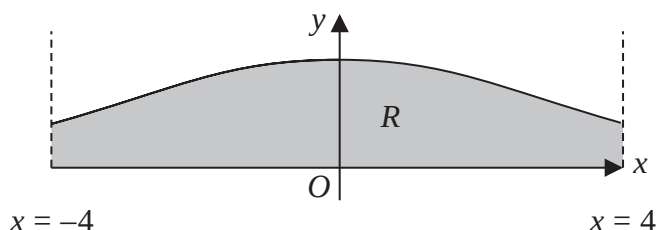


Figure 2

Figure 1 shows a solid paperweight with a flat base.

Figure 2 shows the curve with equation

$$y = H \cos^3 \left(\frac{x}{4} \right) \quad -4 \leq x \leq 4$$

where H is a positive constant and x is in radians.

The region R , shown shaded in Figure 2, is bounded by the curve, the line with equation $x = -4$, the line with equation $x = 4$ and the x -axis.

The paperweight is modelled by the solid of revolution formed when R is rotated 180° about the x -axis.

Given that the maximum height of the paperweight is 2 cm,

(b) write down the value of H . (1)

(c) Using algebraic integration and the result in part (a), determine, in cm^3 , the volume of the paperweight, according to the model. Give your answer to 2 decimal places.

[Solutions based entirely on calculator technology are not acceptable.] (5)

(d) State a limitation of the model. (1)

